K_4 with a Two-Edge Tail: A Rooted Supporting-Plane Proof

Abstract

Let H be obtained from K_4 by attaching an otherwise disjoint path of length two at one clique vertex. We prove

$$t(H, W) \geq p^3(2p-1)(3p-2)$$

for every graphon of edge density $p \in [2/3, 1]$, exactly the chromatic target on its complete required interval. Conditioning at the tail-bearing clique vertex, applying the triangle Goodman bound inside its link, and using the pointwise estimate $\tau \geq (2T_W d - p)_+$ reduce the problem to a scalar function of d and $A = T_W d$. A sharp supporting plane at $(d, A) = (p, p^2)$ integrates exactly because $\int A = \int d^2$. The scalar certificate is proved on the full feasible region; its only lengthy sign is supplied by an explicit positive Bernstein expansion. This proves the formerly open NetworkX Atlas graph 142 without assuming the weighted link inequality that was left unresolved in the failed-attempt ledger.

1 The graph and its exact target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y),$$

and put

$$A(x) = (T_W d)(x) = \int_{\Omega} W(x,y) d(y)\,\mathrm{d}\mu(y).$$

The rooted triangle density is

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Let H have clique vertices o, a, b, c and tail vertices u, v, with all six edges of the K_4 on o, a, b, c and the two further edges ou, uv. After colouring the K_4, each tail vertex has $x-1$ choices. Thus

$$\chi_H(x) = x(x-1)^3(x-2)(x-3), \qquad \chi(H) = 4. \tag{1}$$

For $q = 1 - p > 0$, direct substitution gives

$$\Phi_H(p) = q^6 \chi_H(1/q) = p^3(2p-1)(3p-2). \tag{2}$$

The expression on the right is its polynomial continuation at $p = 1$. The complete required interval is $p \in [2/3, 1]$.

2 The exact link reduction

For $d(x) > 0$, define the probability measure

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y)$$

and let W_x denote the same kernel W on this probability space. Its edge density is

$$z_x = t(K_2, W_x) = \frac{\tau(x)}{d(x)^2}. \tag{3}$$

If $d(x) = 0$, every homomorphism integrand having an edge incident with x vanishes almost everywhere, so that point may simply be assigned zero.

Put, for every $z \geq 0$,

$$f(z) = z(2z-1)_+.$$

The triangle Goodman inequality, combined with nonnegativity, says that every graphon V of edge density z satisfies

$$t(K_3, V) \geq f(z). \tag{4}$$

The function f is nondecreasing: it is zero on $[0, 1/2]$ and has derivative $4z - 1 > 0$ on $(1/2, \infty)$.

The elementary inequality $rs \geq r + s - 1$ on $[0,1]^2$, applied to $W(x,y), W(x,z)$ and multiplied by $W(y,z)$, gives

$$\tau(x) \geq \max\{2A(x) - p, 0\}. \tag{5}$$

Indeed, the two positive terms integrate to $A(x)$ and the negative term to p. Also $\tau(x) \leq d(x)^2$, by dropping the factor $W(y,z)$. Consequently the lower argument used in f below never exceeds the actual link density $z_x \leq 1$.

Let $\kappa_4(x)$ be the rooted K_4 density at $o = x$. Conditioning the other three clique vertices through μ_x gives

$$\kappa_4(x) = d(x)^3 t(K_3, W_x).$$

Integrating the tail vertices first contributes $A(x)$, hence

$$t(H, W) = \int_\Omega d(x)^3 A(x) t(K_3, W_x)\,\mathrm{d}\mu(x). \tag{6}$$

By (4), (5), and monotonicity of f,

$$t(H, W) \geq \int_\Omega \mathcal{F}_p(d(x), A(x))\,\mathrm{d}\mu(x), \tag{7}$$

where

$$\mathcal{F}_p(d,a) = \begin{cases} d^3 a\, f\!\left(\dfrac{(2a-p)_+}{d^2}\right), & d > 0, \\ 0, & d = 0. \end{cases} \tag{8}$$

No pointwise division is made on the zero-degree set.

We shall use the following exact feasible information.

Lemma 2.1 (Rooted feasible region)**.** *For almost every x, with $d = d(x)$ and $a = A(x)$,*

$$0 \leq a \leq d, \qquad a \geq d + p - 1.$$

Moreover,

$$\int_\Omega d\,\mathrm{d}\mu = p, \qquad \int_\Omega A\,\mathrm{d}\mu = \int_\Omega d^2\,\mathrm{d}\mu.$$

Proof. Only the second pointwise inequality needs comment. Applying $rs \geq r+s-1$ to $W(x,y), d(y)$ and integrating gives

$$A(x) \geq \int_\Omega \big(W(x,y) + d(y) - 1\big)\,\mathrm{d}\mu(y) = d(x) + p - 1.$$

The upper bound follows from $d(y) \leq 1$. Finally, symmetry and Tonelli give

$$\int A(x)\,\mathrm{d}\mu(x) = \int W(x,y)d(y)\,\mathrm{d}\mu^2 = \int d(y)^2\,\mathrm{d}\mu(y).$$

The first moment identity is the definition of p. □

3 The sharp scalar supporting plane

Define

$$T_p = p^3(2p-1)(3p-2), \tag{9}$$
$$\gamma_p = p(20p^2 - 15p + 2), \tag{10}$$
$$\beta_p = p^2(30p^2 - 21p + 2), \tag{11}$$

and

$$L_p(d,a) = T_p + \beta_p(d-p) + \gamma_p(a-d^2). \tag{12}$$

The coefficients are the derivatives, in the coordinates $(d, a-d^2)$, of the active expression in (8) at (p,p^2). In particular, $L_p(p,p^2) = T_p$.

Lemma 3.1 (Rooted K_4 supporting plane)**.** *Let $p \in [2/3,1]$ and let $d,a \in [0,1]$ satisfy*

$$0 \leq a \leq d, \qquad a \geq d + p - 1.$$

Then

$$\mathcal{F}_p(d,a) \geq L_p(d,a). \tag{13}$$

Proof. First note that $\gamma_p > 0$ on the stated interval. When $d = 0$, feasibility forces $a = 0$, and

$$L_p(0,0) = -2p^4(12p-7) \leq 0 = \mathcal{F}_p(0,0).$$

Assume henceforth that $d > 0$ and put

$$a_0 = \frac{p}{2} + \frac{d^2}{4}.$$

The scalar function is zero for $a \leq a_0$. At the switching point, direct expansion gives

$$d\big(\mathcal{F}_p(d,a_0) - L_p(d,a_0)\big) = \frac{dp}{4} S_p(d), \tag{14}$$

where

$$\begin{aligned} S_p(d) = {} & 3(20p^2 - 15p + 2)d^2 + (-120p^3 + 84p^2 - 8p)d \\ & + 96p^4 - 96p^3 + 30p^2 - 4p. \end{aligned}$$

Its leading coefficient is positive, and its discriminant in d is

$$-8p(3p-2)^2(120p^3-120p^2+34p-3)\le 0.$$

Indeed, if $u=p-2/3\ge 0$, the final factor is

$$120u^3+120u^2+34u+\frac{17}{9}>0.$$

Thus $S_p(d)\ge 0$. Since L_p is increasing in a, for $a\le a_0$ we obtain

$$L_p(d,a)\le L_p(d,a_0)\le 0=\mathcal{F}_p(d,a),$$

which proves the inactive case.

It remains to take $a\ge a_0$. On this region, put

$$G_{p,d}(a)=d\big(\mathcal{F}_p(d,a)-L_p(d,a)\big). \tag{15}$$

Using $f(z)=z(2z-1)$ here gives the polynomial identity

$$d\mathcal{F}_p(d,a)=2a(2a-p)^2-d^2a(2a-p),$$

so $G_{p,d}$ is a cubic in a with leading coefficient 8. Its discriminant factors exactly as

$$\operatorname{disc}_a G_{p,d}=-4dp(d-p)^2K(p,d). \tag{16}$$

We next give a finite exact positivity certificate for the remaining factor.

For $0\le u,d\le 1$, write

$$B_i^n(s)=\binom{n}{i}s^i(1-s)^{n-i}.$$

With $u=3p-2$, exact coefficient conversion gives

$$K\left(\frac{u+2}{3},d\right)=\frac{1}{765450}\sum_{i=0}^{7}\sum_{j=0}^{6}M_{ij}B_i^7(u)B_j^6(d), \tag{17}$$

where the rows of the positive integer matrix M are

9676800	21504000	24980480	26919060	33180000	48283550	69579300
18835200	50006400	60092480	65235210	79989840	116769575	176444850
33465600	109724800	136834880	149979960	181905520	262135400	403293600
56168640	230669280	299354832	333139662	399385152	560819025	856861470
90512640	469488960	635029632	720982620	856218816	1163172690	1733443740
141264000	931953600	1314243360	1527249330	1803325680	2366134875	3399784650
214617600	1813752000	2664912960	3175640640	3744513360	4758644700	6555605400
318427200	3474122400	5310794160	6495149430	7682260320	9512374725	12564861750

Every Bernstein basis function is nonnegative, and in each coordinate the basis functions sum to one. Hence (17) proves $K(p,d)>0$ throughout $p\in[2/3,1]$, $d\in[0,1]$.

If $d\ne p$, (16) is strictly negative. The cubic $G_{p,d}$ therefore has exactly one real root. Equation (14) is strict in this case: for $p>2/3$ the quadratic S_p has negative discriminant, while at $p=2/3$ one has

$$S_{2/3}(d)=\frac{8}{27}(3d-2)^2,$$

whose only zero is $d = p$. Thus $G_{p,d}(a_0) > 0$. A real cubic with positive leading coefficient and exactly one real root is positive precisely to the right of that root. It follows that $G_{p,d}(a) > 0$ for every $a \geq a_0$.

If $d = p$, direct factorization instead gives

$$G_{p,p}(a) = 2(a - p^2)^2(4a + 7p^2 - 4p).$$

On $a \geq a_0 = p/2 + p^2/4$, the last factor is at least $2p(4p - 1) > 0$. Hence this expression is nonnegative as well. Dividing by $d > 0$ completes the active case and the proof. □

Remark 3.2 (Why this repairs, rather than assumes, the old missing step)**.** *The earlier link approach sought the weighted mean inequality*

$$\int dA\tau \geq \frac{2p - 1}{p} \int d^3 A.$$

That statement does not follow from the accepted degree-weighted lemma, because $A = T_W d$ has no proved monotone ordering along complement edges. Lemma 3.1 never uses that comparison. It keeps the pointwise pair (d, A), applies Goodman before averaging, and uses the exact zero-integral correction $A - d^2$ in its supporting plane.

4 The graphon theorem and Atlas 142

Theorem 4.1 (K_4 with a two-edge tail)**.** *Every graphon W of edge density $p \in [2/3, 1]$ satisfies*

$$\boxed{t(H, W) \geq p^3(2p - 1)(3p - 2) = \Phi_H(p).}$$

Proof. Combine (7) with Lemmas 2.1 and 3.1, and then integrate the supporting plane:

$$\begin{aligned} t(H, W) &\geq \int \mathcal{F}_p(d, A) \\ &\geq T_p + \beta_p \left(\int d - p \right) + \gamma_p \left(\int A - \int d^2 \right) \\ &= T_p. \end{aligned}$$

This is exactly (2). □

At $p = 2/3$, the target is zero and the inequality is also immediate from nonnegativity. At $p = 1$, $W = 1$ almost everywhere and both sides equal one; the proof above remains valid, but this observation handles the endpoint without substituting into $1/(1 - p)$. Every balanced complete k-partite graphon, $k \geq 3$, realizes equality. There $d = p$, $A = p^2$, the link is the balanced complete $(k - 1)$-partite graphon, and every link and scalar bound used above is sharp.

Corollary 4.2 (Atlas 142)**.** *NetworkX Atlas graph* 142*, with graph6 code* `E~AG`*, is positive. It has order six, size eight, chromatic number four, edge list*

$$01,\ 02,\ 03,\ 05,\ 12,\ 13,\ 23,\ 45,$$

and chromatic polynomial

$$x(x - 1)^3(x - 2)(x - 3).$$

Proof. Vertices $0, 1, 2, 3$ induce K_4, and $0 - 5 - 4$ is an otherwise disjoint two-edge tail. Thus this is exactly H. □

All integrands in the proof are bounded, nonnegative, and measurable, so Tonelli–Fubini applies on an arbitrary probability space. No atomlessness, regularity, finite-step reduction, injective-copy interpretation, minimum degree, or pointwise lower degree is used. The only conditional quotient is on $\{d > 0\}$, and the zero-degree set is handled separately.

The theorem is specific to one two-edge tail attached at a K_4 vertex. It does not prove a general clique-tail family, multiple tails, arbitrary attachments to cliques, or the discarded weighted link inequality above. For formal verification, the reusable analytic unit is Lemma 3.1; the graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph142.lean`.

The chromatic identity, rooted algebra, scalar factorizations, discriminant, Bernstein conversion, and Atlas identification were reconstructed in exact rational arithmetic.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/K4Tail/{Scalar,Link,Rows}.lean`, ending at `satisfiesLowerBound_142`. Theorem 4.1 is proved in full, on the whole interval $p \in [2/3, 1]$, with no extra hypothesis. The departure is entirely inside Lemma 3.1.

The departure: neither the cubic discriminant nor the Bernstein matrix is formalized. This note proves Lemma 3.1 by treating $G_{p,d}(a) = d(\mathcal{F}_p(d, a) - L_p(d, a))$ as a cubic in a, factoring $\operatorname{disc}_a G_{p,d} = -4dp(d-p)^2 K(p, d)$, certifying $K > 0$ through the degree-$(7, 6)$ Bernstein expansion (17) with its 8×7 integer matrix, and then invoking the fact that a real cubic with positive leading coefficient and exactly one real root is positive to the right of that root. None of those three ingredients appears in the formalization: not the discriminant, not the matrix M, and not the root-counting argument, which has no convenient Mathlib interface.

Instead the cubic is factored *explicitly*. In the coordinate

$$w = 4a - 2p - d^2, \qquad \text{so that} \qquad d\,\mathcal{F}_p(d, a) = \tfrac{1}{2}\, a\, w\, (w + d^2),$$

one has the polynomial identity, checked by `ring`,

$$\begin{aligned}(11p-4)^3 \cdot 8 \cdot \big(a(2a-p)w - d\,L_p(d,a)\big) = \big((11p-4)w - R\big)^2\big((11p-4)w + 2R + 2(11p-4)(p+d^2)\big) \\ + (11p-4)(d-p)^2\,\Lambda\, w + (d-p)^2\,\Gamma,\end{aligned} \tag{18}$$

where $R = (11p-4)r$ and $r(p, d) = \big(3p^2 - 2p\big) + \sigma(p)(d-p)$ is the tangent line, at $d = p$, of the double root of the cubic; the slope $\sigma(p) = (6p^2 - 9p + 2)/(11p - 4)$ is forced, and is the reason for the factor $11p - 4$. The two remainder coefficients are

$$\begin{aligned}\Lambda &= \big((11p-4)d + 23p^2 - 22p + 4\big)^2 + \big(1658p^4 - 1850p^3 + 647p^2 - 60p - 4\big), \\ \Gamma &= Ad^2 + Bd + C, \qquad A = -2(11p-4)(6p^2 - 9p + 2)^2 \le 0.\end{aligned}$$

$\Lambda \ge 0$ is a square plus a quartic in p alone. For Γ, the feasibility $a \le d$ together with $w \ge 0$ gives $d^2 \le 4d - 2p$, and since $A \le 0$ this yields $\Gamma \ge (4A + B)d + C - 2Ap$; the same feasibility gives $d \ge p/2$, and $4A + B \ge 0$ then leaves $\Gamma \ge \frac{1}{2}Bp + C$, again a polynomial in p alone.

Every one-variable inequality that remains — there are seven — has *all* coefficients nonnegative after $p = \frac{2}{3} + u$, so each is a one-line `nlinarith`. In particular the quantity $1658p^4 - 1850p^3 +$

$647p^2 - 60p - 4$, whose positivity replaces the discriminant condition of this note, is $\frac{1856}{81} + \frac{8128}{27}u + \frac{4105}{3}u^2 + \frac{7714}{3}u^3 + 1658u^4$.

The two routes prove the same lemma on the same interval. The note's is the more informative one — it exhibits the root structure — while (18) is the one a proof assistant can check without a theory of cubics.

The truncations are carried, not divided away. The note defines $\mathcal{F}_p(d, 0) = 0$ by hand and notes that no pointwise division is made on the zero-degree set. The formalization avoids the definition entirely: `K4Tail.mul_rootedK4_ge` states the link bound in the form

$$d(x)\,\kappa_4(x) \;\geq\; u \cdot (2u - d(x)^2)_+, \qquad u = (2A(x) - p)_+,$$

multiplied through by $d(x)$, so both sides are polynomials in the truncated quantities and $d(x) = 0$ is an ordinary case rather than a convention. The monotonicity of f used in (7) becomes the elementary step that $t \mapsto t(2t - d^2)$ is nondecreasing once $2t \geq d^2$, applied to $\tau \geq u$.

Everything else transcribes directly. (4) is `K4Tail.goodman`, the pure-chordal clique bound at $r = 3$ extended below the threshold by nonnegativity; (6) is the existing cone identity `Link.rootedDensity_eq` at $F = K_3$; (5) is `Link.rootedTriangle_ge`; and the two moment identities of Lemma 2.1 are `Foundation.integral_degree` and `BookTail.integral_pathOp`. The final integration of the supporting plane is `K4Tail.integral_plane`, which is exactly the statement that the two corrections $d - p$ and $A - d^2$ have mean zero.